CORRECTION NEUROPHYSIOLOGY MIDTERM, PRACTICE 1

Correct answers in red, with a short explanation.

1- After leaving the fourth ventricle, cerebrospinal fluid reaches the subarachnoid space mainly through

A- The cerebral aqueduct
B- The choroid plexus of the lateral ventricle
C- The arachnoid granulations
D- The foramen of Magendie
E- The foramen of Monro

Answer D. CSF flow: lateral ventricles → foramen of Monro → third ventricle → cerebral aqueduct → fourth ventricle → foramen of Magendie → subarachnoid space → arachnoid granulations → superior sagittal sinus.

2- Which sympathetic target receives postganglionic fibres that release acetylcholine?

A- The sinoatrial node
B- Bronchial smooth muscle
C- The dilator muscle of the pupil
D- Sweat glands
E- Renal blood vessels

Answer D. Sweat glands are the cholinergic sympathetic exception. Renal vessels receive dopamine; the adrenal medulla is reached by preganglionic fibres.

3- A neurologist places a vibrating tuning fork on a patient's fingertip. The receptor mainly responsible for detecting the vibration is the

A- Free nerve ending
B- Pacinian corpuscle
C- Merkel disc
D- Ruffini ending
E- Meissner corpuscle

Answer B. Pacinian corpuscles are rapidly adapting vibration detectors. Merkel: form and texture; Meissner: motion and grip; Ruffini: skin stretch.

4- A man touches a hot pan. He first feels a sharp, well-localised pain, followed a second later by a dull burning pain. The burning second pain is carried by

A- Thinly myelinated A-delta fibres
B- A-alpha fibres from muscle spindles
C- Large myelinated A-beta fibres
D- Unmyelinated C fibres
E- Gamma motor fibres

Answer D. First (sharp) pain travels in A-delta fibres; second (burning) pain in slow unmyelinated C fibres.

5- The dorsal (posterior) column pathway carries

A- Visceral pain from the abdominal organs
B- Pain, temperature and crude touch
C- Automatic postural tone from the vestibular nuclei
D- Voluntary motor commands to distal limbs
E- Discriminative touch, vibration and conscious proprioception

Answer E. Pain, temperature and crude touch travel in the anterolateral (spinothalamic) system.

6- Thanks to the buoyancy provided by cerebrospinal fluid, the effective weight of a 1400 g brain is reduced to about

A- 5 g
B- 700 g
C- 350 g
D- 50 g
E- 150 g

Answer D. Buoyancy is one of the four functions of CSF, with protection, excretion of waste and acting as an endocrine medium.

7- At rest, the membrane potential of a neuron (about -70 mV) lies closest to the equilibrium potential of

A- K^+, because K^+ leak channels are open
B- Na^+, because the Na^+/K^+ pump moves 3 Na^+ out
C- Ca^{2+}, because Ca^{2+} is high outside
D- Cl^-, because Cl^- is actively pumped in
E- Na^+, because Na^+ leak channels are open

Answer A. At rest K^+ permeability is high, so the potential sits near EK (about -90 mV); few Na^+ channels are open.

8- The equilibrium potential of Na^+ in a typical neuron is approximately

A- -70 mV
B- +60 mV
C- -90 mV
D- -55 mV
E- 0 mV

Answer B. ENa about +60 mV; EK about -90 mV; rest -70 mV; threshold -50 to -55 mV.

9- A toxin removes the inactivation gate of voltage-gated Na^+ channels but leaves activation intact. The most likely effect on the action potential is

A- A larger afterhyperpolarization only
B- Loss of the rising phase
C- A faster repolarization
D- A more negative resting potential
E- A prolonged depolarization, because Na^+ keeps entering

Answer E. Inactivation is what stops Na^+ entry and allows the falling phase; without it the membrane stays depolarized.

10- Saltatory conduction is possible because voltage-gated channels are concentrated at the

A- Dendritic spines
B- Cell body
C- Nodes of Ranvier
D- Internodes beneath the myelin
E- Axon terminals only

Answer C. Myelin (up to 300 layers) has no channels beneath it, so the signal jumps from node to node.

11- In a typical multipolar neuron, the action potential is initiated at the

A- Axon terminal
B- Axon hillock
C- Node of Ranvier nearest the terminal
D- Cell body nucleus
E- Dendrites

Answer B. Trigger zone = axon hillock. Input zone = dendrites and soma; output zone = axon terminals.

12- Which statement correctly describes a graded potential?

A- It always depolarizes the membrane
B- It is all-or-none
C- Its amplitude is proportional to the stimulus and it decreases with distance
D- It travels long distances without decrement
E- It occurs only in membranes with voltage-gated channels

Answer C. Graded potentials are local and can be excitatory or inhibitory; action potentials are all-or-none and travel without decrement.

13- The afterhyperpolarization at the end of an action potential is mainly caused by

A- Closure of K^+ leak channels
B- Entry of Ca^{2+} through ligand-gated channels
C- Slow closure of voltage-gated K^+ channels
D- Activation of the Na^+/K^+ pump alone
E- Opening of voltage-gated Na^+ channels

Answer C. K^+ channels open and close slowly, so K^+ keeps leaving briefly after the membrane is back at rest.

14- A drug blocks voltage-gated Ca^{2+} channels in the presynaptic terminal. The action potential still reaches the terminal normally. What is the most likely result?

A- Neurotransmitter reuptake is increased
B- Acetylcholinesterase is inhibited
C- The action potential can no longer propagate along the axon
D- Postsynaptic receptors are blocked
E- Neurotransmitter release is reduced

Answer E. Ca^{2+} entry is the link between the electrical and the chemical step: it triggers exocytosis of synaptic vesicles.

15- The main inhibitory neurotransmitter in the brain is

A- GABA
B- Glutamate
C- Noradrenaline
D- Acetylcholine
E- Substance P

Answer A. Glutamate is the main excitatory transmitter; GABA (and glycine) are inhibitory.

16- An inhibitory postsynaptic potential (IPSP) is typically produced by

A- Entry of Cl^-, which hyperpolarizes the membrane
B- Entry of Ca^{2+} into the presynaptic terminal
C- Closure of K^+ leak channels
D- Entry of Na^+, which depolarizes the membrane
E- Activation of voltage-gated Na^+ channels

Answer A. EPSP: positive ions in (Na^+), towards threshold. IPSP: negative ions in (Cl^-), away from threshold.

17- A single presynaptic neuron fires three action potentials in rapid succession at the same synapse, and the postsynaptic neuron reaches threshold. This is an example of

A- Temporal summation
B- Habituation
C- Spatial summation
D- Divergence
E- Lateral inhibition

Answer A. Temporal: one neuron, repeated firing. Spatial: several neurons at the same time at different places. Each PSP is only about 10 mV, so summation is needed.

18- Selective serotonin reuptake inhibitors (SSRIs) act by

A- Keeping serotonin longer in the synaptic cleft
B- Converting serotonin into GABA
C- Blocking serotonin receptors
D- Increasing serotonin breakdown by enzymes
E- Preventing serotonin vesicle release

Answer A. Blocking reuptake prolongs the transmitter's action, the lecture's example of a drug acting on removal.

19- You no longer feel your watch on your wrist a few minutes after putting it on. This is mainly due to

A- Sensitization by serotonin
B- Rapid adaptation of phasic touch receptors
C- Damage to the dorsal column
D- Slow adaptation of tonic proprioceptors
E- Lateral inhibition in the thalamus

Answer B. Phasic receptors (touch, temperature, smell) signal change and adapt rapidly; tonic receptors (proprioceptors) adapt slowly.

20- A patient has a hemisection of the right half of the spinal cord at T10. Which finding is expected below the lesion?

A- Loss of vibration sense on the left side
B- Loss of pain and temperature on the right side
C- Loss of all sensation on both sides
D- Loss of pain and temperature on the left side
E- Loss of fine touch on the left side

Answer D. Pain and temperature fibres cross in the spinal cord, so they are lost on the opposite side. Fine touch, vibration and proprioception cross in the medulla, so they are lost on the same side.

21- In smooth muscle, the Ca^{2+}-binding protein that starts contraction is

A- Titin
B- Calmodulin
C- Tropomyosin
D- Nebulin
E- Troponin

Answer B. Skeletal and cardiac muscle use troponin; smooth muscle uses calmodulin.

22- During contraction of a skeletal muscle fibre, which region keeps a constant length?

A- The H zone
B- The sarcomere
C- The A band
D- The I band
E- The distance between Z discs

Answer C. The A band equals the thick filament length, and filaments do not change length. The I band and H zone shorten.

23- A triad in skeletal muscle is made of

A- Three myofibrils bound by titin
B- Two T-tubules and one terminal cisterna
C- Actin, myosin and troponin
D- One Z disc and two M lines
E- One T-tubule and two terminal cisternae of the sarcoplasmic reticulum

Answer E. The T-tubule carries the action potential; the terminal cisternae store and release Ca^{2+}.

24- A muscle tissue is striated, involuntary and its cells are linked by gap junctions. It is

A- Skeletal muscle
B- Multi-unit smooth muscle
C- Cardiac muscle
D- Myoepithelium
E- Single-unit smooth muscle

Answer C. Smooth muscle is not striated; skeletal muscle is voluntary and has no gap junctions. Cardiac cells connect through intercalated discs.

25- A student holds a heavy suitcase still with the elbow flexed at 90°. The biceps develops tension without changing length. This is

A- A passive stretch

B- A tetanic shortening

C- An isotonic eccentric contraction

D- An isotonic concentric contraction

E- An isometric contraction

Answer E. Isometric: myosin heads generate tension but the thin filaments cannot slide because the load is too great. Isotonic: the muscle shortens and the load moves.